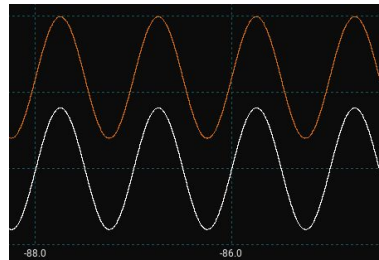


Dynamic Self-Synchronizing Signals System

Q1: Engineering Problem & Objectives

- Present synchronization methods has limitations:
 - Complex Logic
 - Introduce Interferences
 - Modify the Phase of the Signals
- Objective: Find a method to synchronize signals
 - Low-Cost & Simple Logic
 - High Precision
 - Low-Amplitude Sensitivity
 - Less Interference
 - Preserve Signal Integrity

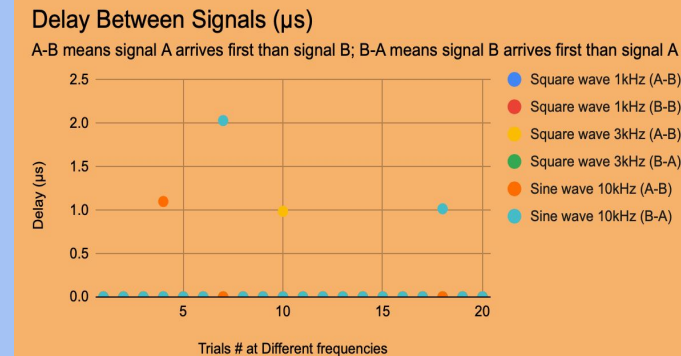
Figure 1: Synchronized Signals.
Image Created By Student using
PicoScope 7 T&M, 2026



Q3: Data Analysis & Results

- Analysis
 - Scope Precision: **<4ns**
 - **"0s" = <4ns**
- Results:
 - **4ns < precision**
 - **96.67% reliability**
 - **Costs \$2.97**
(producing 1)

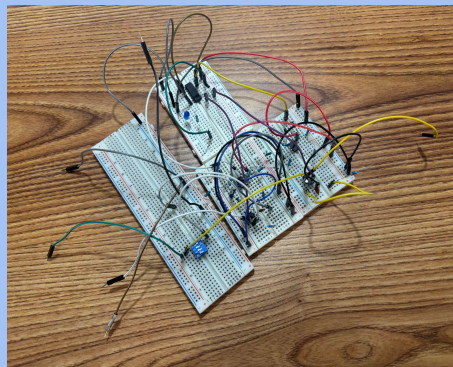
Figure 3: Analyzed Datasets
Graph Created By Student
using Google Sheets, 2026



Q2: Project Design:

- **Logic:**
 - Signals Arrive
 - Generates Logic-high Voltages
 - Detects All High Voltage
 - Turn On Switch
 - Signals Pass to the Backend Device

Figure 2: My prototype.
Image created by student
using camera, 2026



Q4: Interpretations and Conclusion

- **Applications:**
 - Automotive Device
 - Medical Device
 - Industrial Automation Device
- **Theoretical Outcomes:**
 - 10ns delay maximum
 - Bandwidth: 60kHz
- **Challenges:**
 - Parasitic Effects on the breadboard
 - Lack of oscilloscope precision
 - Limited components bandwidth and precision